

Bellabeat SQL Log

1. Data Validation

Verified that all four cleaned datasets were successfully imported into BigQuery. Confirmed table schemas, column names, and data types. Date/time fields were imported as DATE or TIMESTAMP where appropriate, and identifier fields were imported as numeric or string values according to their intended use.

<u>Table</u>	<u>Purpose</u>
daily_activity	Daily activity, steps, distance, active minutes, sedentary time, and calories
hourly_intensities	Activity intensity patterns by hour
sleep_day	Sleep duration and time spent in bed
weight_log_info	Weight, BMI, and weight-tracking information

2. Analytical Queries

2.1- Average Activity by User

Purpose: Establish the average daily activity level for each user and identify variation in activity across the sample.

Analytical Question:How active are individual users on average, and how much does activity vary between users?

SQL

SELECT

```
Id,

COUNT(*) AS days_recorded,

ROUND(AVG(TotalSteps), 0) AS avg_daily_steps,

ROUND(AVG(VeryActiveMinutes), 1) AS avg_very_active_minutes,

ROUND(AVG(FairlyActiveMinutes), 1) AS avg_fairly_active_minutes,
```

```

ROUND(AVG(LightlyActiveMinutes), 1) AS avg_lightly_active_minutes,

ROUND(AVG(SedentaryMinutes), 1) AS avg_sedentary_minutes

FROM `bellabeat-analysis-506320.bellabeat_data.daily_activity`

GROUP BY Id

ORDER BY avg_daily_steps DESC;

```

Key Result: The analysis included 35 users. Average daily steps ranged from 774 to 16,759 steps per day, indicating substantial variation in activity levels across the sample.

Interpretation: The large difference in average activity suggests that users exhibit meaningfully different activity behaviors. Further analysis is needed to determine what characteristics or behaviors may distinguish lower-activity users from more-active users.

2.2- User Activity Distribution

Purpose: Quantify the distribution of activity levels among users and determine how prevalent lower activity levels are within the sample.

Analytical Question: How widespread are lower activity levels among the users?

SQL

SELECT

```

ROUND(AVG(avg_daily_steps), 0) AS overall_avg_steps,

ROUND(MIN(avg_daily_steps), 0) AS lowest_user_avg,

ROUND(MAX(avg_daily_steps), 0) AS highest_user_avg,

ROUND(AVG(CASE WHEN avg_daily_steps < 5000 THEN avg_daily_steps END), 0) AS
avg_steps_low_activity_users,

COUNTIF(avg_daily_steps < 5000) AS users_below_5000_steps

FROM (

SELECT

    Id,

```

```

    AVG(TotalSteps) AS avg_daily_steps

FROM `bellabeat-analysis-506320.bellabeat_data.daily_activity`

GROUP BY Id

);

```

Key Result: The average user recorded 7,079 steps per day. Eleven of the 35 users averaged fewer than 5,000 steps per day. These users averaged 2,898 steps per day.

Interpretation: Lower activity was not limited to a small number of users. Approximately 31.4% of the sample averaged fewer than 5,000 steps per day, suggesting that lower activity may represent a broader behavioral pattern worth investigating.

2.3- Activity Profile Comparison

Purpose: Compare activity characteristics between lower- and higher-activity users to identify behavioral differences that may explain variation in daily steps.

Analytical Question: How do activity patterns differ between users averaging fewer than 5,000 steps per day and those averaging 5,000 or more?

SQL:

SELECT

```

CASE

    WHEN avg_daily_steps < 5000 THEN 'Lower Activity'

    ELSE 'Higher Activity'

END AS activity_group,

COUNT(*) AS users,

ROUND(AVG(avg_daily_steps), 0) AS avg_daily_steps,

ROUND(AVG(avg_very_active_minutes), 1) AS avg_very_active_minutes,

ROUND(AVG(avg_fairly_active_minutes), 1) AS avg_fairly_active_minutes,

ROUND(AVG(avg_lightly_active_minutes), 1) AS avg_lightly_active_minutes,

ROUND(AVG(avg_sedentary_minutes), 1) AS avg_sedentary_minutes

```

```

FROM (

SELECT

    Id,

    AVG(TotalSteps) AS avg_daily_steps,

    AVG(VeryActiveMinutes) AS avg_very_active_minutes,

    AVG(FairlyActiveMinutes) AS avg_fairly_active_minutes,

    AVG(LightlyActiveMinutes) AS avg_lightly_active_minutes,

    AVG(SedentaryMinutes) AS avg_sedentary_minutes

FROM `bellabeat-analysis-506320.bellabeat_data.daily_activity`

GROUP BY Id

)

GROUP BY activity_group

ORDER BY avg_daily_steps;

```

Key Result: Lower-activity users averaged 2,898 steps per day compared with 8,995 among higher-activity users. Lower-activity users also recorded substantially fewer very active minutes (3.5 vs. 26.2) and lightly active minutes (117.2 vs. 216.0), while averaging more sedentary minutes (1,171.2 vs. 939.6).

Interpretation: Lower-activity users appear to have a less active overall profile, characterized by substantially less vigorous and light activity and more sedentary time. Further analysis is needed to determine whether sedentary behavior, lack of higher-intensity activity, or another behavioral factor is most closely associated with lower daily activity.

2.4- Sleep Patterns by Activity Level

Purpose: Determine whether sleep duration and time spent in bed differ between lower- and higher-activity users.

Analytical Question: Do lower-activity users differ from higher-activity users in average sleep duration or time spent in bed?

SQL

```
SELECT

    activity_group,

    COUNT(*) AS users,

    ROUND(AVG(avg_daily_steps), 0) AS avg_daily_steps,

    ROUND(AVG(avg_minutes_asleep), 1) AS avg_minutes_asleep,

    ROUND(AVG(avg_minutes_in_bed), 1) AS avg_minutes_in_bed

FROM (

    SELECT

        activity.Id,

        CASE

            WHEN activity.avg_daily_steps < 5000 THEN 'Lower Activity'

            ELSE 'Higher Activity'

        END AS activity_group,

        activity.avg_daily_steps,

        sleep.avg_minutes_asleep,

        sleep.avg_minutes_in_bed

    FROM (

        SELECT

            Id,

            AVG(TotalSteps) AS avg_daily_steps

        FROM `bellabeat-analysis-506320.bellabeat_data.daily_activity`

        GROUP BY Id
```

```

) AS activity

INNER JOIN (

SELECT

    Id,

    AVG(TotalMinutesAsleep) AS avg_minutes_asleep,

    AVG(TotalTimeInBed) AS avg_minutes_in_bed

FROM `bellabeat-analysis-506320.bellabeat_data.sleep_day`

GROUP BY Id

) AS sleep

ON activity.Id = sleep.Id

)

GROUP BY activity_group

ORDER BY avg_daily_steps;

```

Key Result: Among the 24 users with both activity and sleep data, average sleep duration was nearly identical between lower-activity users (378.6 minutes) and higher-activity users (377.0 minutes). However, lower-activity users spent approximately 30 more minutes in bed on average (441.0 vs. 411.2 minutes).

Interpretation: Sleep duration did not appear to differ meaningfully between the two activity groups. However, lower-activity users spent more time in bed without a corresponding increase in sleep duration. This pattern warrants further investigation.

2.5- Peak Activity Timing

Purpose: Identify the times of day when users most commonly reach their highest activity intensity and determine whether the overall daily activity pattern represents a consistent behavior across individual users.

Analytical Question: When do individual users tend to reach their peak activity intensity?

SQL

```
WITH hourly_user_activity AS (  
  
    SELECT  
  
        Id,  
  
        EXTRACT(HOUR FROM ActivityHour) AS hour_of_day,  
  
        AVG(AverageIntensity) AS avg_hourly_intensity  
  
    FROM `bellabeat-analysis-506320.bellabeat_data.hourly_intensities`  
  
    GROUP BY Id, hour_of_day  
  
) ,
```

```
ranked_hours AS (  
  
    SELECT  
  
        Id,  
  
        hour_of_day,  
  
        avg_hourly_intensity,  
  
        RANK() OVER (  
  
            PARTITION BY Id  
  
            ORDER BY avg_hourly_intensity DESC  
  
        ) AS hour_rank  
  
    FROM hourly_user_activity  
  
)
```

```
SELECT  
  
    hour_of_day,
```

```

COUNT(*) AS users_with_peak_activity

FROM ranked_hours

WHERE hour_rank = 1

GROUP BY hour_of_day

ORDER BY users_with_peak_activity DESC;

```

Key Result: Overall hourly activity intensity was highest around 7 PM. However, analysis at the individual-user level showed that peak activity times were distributed across multiple hours. Noon was the most common individual peak hour, with 8 users, followed by 5 PM with 5 users, 7 PM with 4 users, and 11 AM with 4 users. The analysis included 34 users and 35 peak-hour records because one user had two hours tied for their highest average intensity.

Interpretation: Although overall activity intensity peaked in the evening, individual users did not consistently reach their highest activity at the same time. The distribution of peak activity across several times of day suggests that Bellabeat should avoid assuming a single universal period of peak engagement. However, the concentration of individual peaks around midday and the early evening may provide useful opportunities for time-specific engagement strategies.

2.6- Sustained Activity by Time of Day

Purpose: Determine whether the period of highest activity intensity is consistent across individual users when the day is divided into broader time periods.

Analytical Question: Which period of the day has the highest average activity intensity for the greatest number of users?

SQL

```

WITH user_period_activity AS (

SELECT

    Id,

    CASE

        WHEN EXTRACT(HOUR FROM ActivityHour) BETWEEN 0 AND 5

            THEN 'Overnight'

```



```

        WHEN EXTRACT(HOUR FROM ActivityHour) BETWEEN 6 AND 11

            THEN 'Morning'

        WHEN EXTRACT(HOUR FROM ActivityHour) BETWEEN 12 AND 17

            THEN 'Afternoon'

        WHEN EXTRACT(HOUR FROM ActivityHour) BETWEEN 18 AND 23

            THEN 'Evening'

    END AS time_period,

    AVG(AverageIntensity) AS avg_period_intensity

FROM `bellabeat-analysis-506320.bellabeat_data.hourly_intensities`

GROUP BY Id, time_period

),

ranked_periods AS (

    SELECT

        Id,

        time_period,

        avg_period_intensity,

        RANK() OVER (

            PARTITION BY Id

            ORDER BY avg_period_intensity DESC

        ) AS period_rank

    FROM user_period_activity

)

```

```

SELECT

    time_period,

    COUNT(*) AS users_with_peak_activity

FROM ranked_periods

WHERE period_rank = 1

GROUP BY time_period

ORDER BY users_with_peak_activity DESC;

```

Key Result: Afternoon was the highest-intensity period for 22 of the 34 users with hourly intensity data (64.7%). Evening and morning were each the highest-intensity period for 6 users (17.6%), while no users had overnight as their highest-intensity period.

Interpretation: Afternoon activity was consistently elevated across the majority of users, suggesting that the afternoon may represent a particularly strong opportunity for Bellabeat to engage users through activity reminders, challenges, or other interventions. The consistency across individual users strengthens the finding beyond the overall sample-level average.

2.7- Daily Activity Composition by Activity Level

Purpose: Compare how lower- and higher-activity users distribute their daily time across sedentary, lightly active, fairly active, and very active behaviors.

Analytical Question: How does the composition of daily activity differ between lower- and higher-activity users?

SQL

```

WITH user_activity AS (

    SELECT

        Id,

        AVG(TotalSteps) AS avg_daily_steps,

        AVG(SedentaryMinutes) AS avg_sedentary_minutes,

```

```

    AVG(LightlyActiveMinutes) AS avg_lightly_active_minutes,

    AVG(FairlyActiveMinutes) AS avg_fairly_active_minutes,

    AVG(VeryActiveMinutes) AS avg_very_active_minutes

FROM `bellabeat-analysis-506320.bellabeat_data.daily_activity`

GROUP BY Id
)

SELECT

CASE

    WHEN avg_daily_steps < 5000 THEN 'Lower Activity'

    ELSE 'Higher Activity'

END AS activity_group,

COUNT(*) AS users,

ROUND(AVG(avg_sedentary_minutes) / 1440 * 100, 1) AS sedentary_pct,

ROUND(AVG(avg_lightly_active_minutes) / 1440 * 100, 1) AS lightly_active_pct,

ROUND(AVG(avg_fairly_active_minutes) / 1440 * 100, 1) AS fairly_active_pct,

ROUND(AVG(avg_very_active_minutes) / 1440 * 100, 1) AS very_active_pct

FROM user_activity

GROUP BY activity_group

ORDER BY

CASE activity_group

    WHEN 'Lower Activity' THEN 1

    WHEN 'Higher Activity' THEN 2

```

```
END;
```

Key Result: Lower-activity users spent an average of 81.3% of the recorded day sedentary, compared with 65.2% among higher-activity users. Higher-activity users spent a greater proportion of the day lightly active (15.0% vs. 8.1%), fairly active (1.1% vs. 0.8%), and very active (1.8% vs. 0.2%).

Interpretation: The primary difference between lower- and higher-activity users was not limited to periods of intense exercise. Higher-activity users spent substantially less time sedentary and a greater proportion of their day engaged in activity, particularly light activity. This suggests that Bellabeat may have an opportunity to encourage more frequent movement throughout the day rather than focusing exclusively on high-intensity exercise.

2.8- Consistency of Daily Activity

Purpose: Determine whether higher-activity users maintain their activity more consistently than lower-activity users.

Analytical Question: Are higher-activity users consistently active, or are their higher average step counts driven by occasional unusually active days?

Method: Daily step variability was evaluated at the user level using the coefficient of variation (CV), calculated as each user's standard deviation of daily steps divided by their average daily steps. CV was used instead of raw step range because it measures variability relative to each user's typical activity level, allowing for a more meaningful comparison between lower- and higher-activity users.

SQL

```
WITH user_activity AS (  
  
  SELECT  
  
    Id,  
  
    AVG(TotalSteps) AS avg_daily_steps,  
  
    STDDEV(TotalSteps) AS stddev_daily_steps  
  
  FROM `bellabeat-analysis-506320.bellabeat_data.daily_activity`  
  
  GROUP BY Id  
  
) ,
```

```

user_consistency AS (

    SELECT

        Id,

        avg_daily_steps,

        stddev_daily_steps,

        SAFE_DIVIDE(stddev_daily_steps, avg_daily_steps) AS coefficient_of_variation

    FROM user_activity

    WHERE avg_daily_steps > 0

)

```

```

SELECT

    CASE

        WHEN avg_daily_steps < 5000 THEN 'Lower Activity'

        ELSE 'Higher Activity'

    END AS activity_group,

    COUNT(*) AS users,

    ROUND(AVG(avg_daily_steps), 0) AS avg_daily_steps,

    ROUND(AVG(stddev_daily_steps), 0) AS avg_stddev,

    ROUND(AVG(coefficient_of_variation) * 100, 1) AS avg_cv_pct

FROM user_consistency

GROUP BY activity_group

ORDER BY

```

```
CASE activity_group

  WHEN 'Lower Activity' THEN 1

  ELSE 2

END;
```

Key Result: Lower-activity users had an average coefficient of variation of **115.1%**, compared with **46.5%** among higher-activity users. A median-CV validation produced a similar result (**110.0% vs. 44.2%**), indicating that the difference was not primarily driven by a small number of extreme users.

Interpretation: Higher-activity users demonstrated substantially less day-to-day variation in steps relative to their typical activity level. This suggests that higher activity in this sample is associated not only with taking more steps, but also with maintaining a more consistent daily activity pattern.

Limitation: The analysis identifies an association rather than a causal relationship. The lower-activity group also contained fewer users (11 vs. 24), so the difference should be interpreted cautiously.

2.9- Frequency of Very Low Activity Days

Purpose: Determine whether lower-activity users experience very-low-activity days more frequently than higher-activity users.

Analytical Question: Do lower-activity users experience a greater frequency of very-low-activity days, rather than simply having a lower average number of daily steps?

Method: Users were classified into the existing Lower Activity and Higher Activity groups based on average daily steps, using 5,000 steps as the group threshold. A very-low-activity day was initially defined as a day with fewer than 2,000 steps. For each user, the percentage of recorded days below the threshold was calculated. Average and median percentages were then compared between activity groups. A second analysis using a 3,000-step threshold was performed as a sensitivity check to determine whether the observed pattern depended heavily on the initial cutoff.

SQL- Primary Analysis

```
WITH user_days AS (

  SELECT
```

```

    Id,

    AVG(TotalSteps) AS avg_daily_steps,

    COUNTIF(TotalSteps < 2000) AS low_activity_days,

    COUNT(*) AS total_days

FROM `bellabeat-analysis-506320.bellabeat_data.daily_activity`

GROUP BY Id

),

user_low_activity AS (

    SELECT

        Id,

        avg_daily_steps,

        low_activity_days,

        total_days,

        SAFE_DIVIDE(low_activity_days, total_days) * 100 AS low_activity_pct

    FROM user_days

)

SELECT

    CASE

        WHEN avg_daily_steps < 5000 THEN 'Lower Activity'

        ELSE 'Higher Activity'

    END AS activity_group,

```

```

COUNT(*) AS users,

ROUND(AVG(low_activity_days), 1) AS avg_low_activity_days,

ROUND(AVG(low_activity_pct), 1) AS avg_low_activity_pct

FROM user_low_activity

GROUP BY activity_group

ORDER BY

CASE activity_group

    WHEN 'Lower Activity' THEN 1

    ELSE 2

END;

```

Primary Results:

Activity Group	Users	Avg. Low-Activity Days	Avg. % of Recorded Days <2,000 Steps
Lower Activity	11	15.2	47.8%
Higher Activity	24	3.0	7.4%

Sensitivity Check- 3,000-Step Threshold

A second analysis using fewer than 3,000 steps as the definition of a very-low-activity day produced:

Activity Group	Users	Avg. Low-Activity Days	Avg. % of Recorded Days <3,000 Steps
Lower Activity	11	18.8	58.2%
Higher Activity	24	4.8	11.7%

The difference between the groups remained substantial when the threshold was increased from 2,000 to 3,000 steps, indicating that the observed pattern was not dependent on a single arbitrary cutoff.

Median Validation - 2,000-Step Threshold

A median comparison was also performed to determine whether the primary result was being driven by a small number of users with unusually high percentages of very-low-activity days.

Activity Group	Users	Avg. % <2,000 Steps	Median % <2,000 Steps
Lower Activity	11	47.8%	54.5%
Higher Activity	24	7.4%	4.9%

The median results supported the primary analysis and showed that the difference was not simply caused by a few extreme users.

Key Result: Lower-activity users experienced very-low-activity days substantially more frequently than higher-activity users. At the 2,000-step threshold, lower-activity users averaged 47.8% of recorded days below the threshold, compared with 7.4% for higher-activity users. The median comparison showed an even larger distinction: 54.5% vs. 4.9%.

Interpretation: Lower activity in this sample appears to involve not only a lower average number of steps but also a much higher frequency of days with extremely low movement. This complements the finding from 2.8, which showed that higher-activity users had substantially lower relative day-to-day variability. Together, the findings suggest that consistency and reducing periods of very low movement may be important behavioral opportunities for Bellabeat.

Business Implication: Bellabeat could potentially focus engagement strategies on helping lower-activity users avoid extended periods of very low movement. Rather than emphasizing intense exercise alone, interventions could encourage users to maintain a minimum level of daily movement through simple, achievable activity prompts.

Limitations: The 2,000- and 3,000-step thresholds were used as exploratory analytical cutoffs rather than established health thresholds. The analysis is observational and does not establish that reducing very-low-activity days would cause users to become more active. The Lower Activity group also contained fewer users, 11, than the Higher Activity group, 24, so results should be interpreted with appropriate caution.